

# Supply Chain Data Analysis Report

## 1. Problem Statement:

The Objective of this project is to analyze supply chain data to understand order performance, product demand, supplier performance delivery performance, and sales-related matrices.

The project aims to convert raw supply chain data into meaningful information that can help management make better decisions related to inventory, suppliers, deliveries and overall supply chain performance.

## 2. Business Requirements:

- Identify the total number of orders.
- Analyze the total ordered and delivered quantities.
- Identify products with the highest delivered quantity.
- Calculate On-Time In-Full (OTIF) Performance.
- Analyze product performance by category.
- Compare ordered quantity with delivered quantity.
- Provide meaningful insights that can support business decision – making.
- Identify products or suppliers that may require improvement.

## 3. Tools Used:

- Data exploration
- Data cleaning and validation
- Filtering and sorting data
- Aggregation
- Group by and Having
- Inner Join and Left Join
- KPI Calculation
- **Power BI / Excel:**
  - Data Visualization
  - KPI dashboards
  - Charts and graphs
  - Trend analysis
  - Interactive reporting
- **SQL Techniques**
  - **Select**
  - **WHERE**
  - **Group By**
  - **HAVING**
  - **Order By**
  - **Aggregate Functions**

- **Inner Join**
- **Left Join**

#### **4. Analysis:**

##### ➤ **Order Analysis:**

The total number of orders was analyzed to understand the overall volume of business managed by the supply chain.

The number of was also analyzed across various products, suppliers, and regions to identify areas with higher and lower activity.

##### ➤ **Product Analysis:**

Product-level analysis was performed using ordered and delivered quantities. The analysis helps identify.

- High – demand products
- Low-demand products
- Products with high delivered quantities
- Products with delivery shortages

This information can help the business improve inventory planning and product availability.

##### ➤ **Supplier Analysis:**

Supplier performance was analyzed based on the number of orders handled by each supplier.

Supplier with a higher number of orders were identified, and supplier – level performance can be further evaluated using:

- Number of orders
- Ordered quantity.
- Delivered quantity.
- On-time delivery.
- In-full delivery.

##### ➤ **Delivery Analysis:**

Delivery performance was analyzed by comparing the ordered quantity with the delivery quantity.

This analysis helps identify potential supply shortages and fulfillment problems

##### ➤ **Product Category Analysis**

- Number of Orders per category
- Average unit price per category
- Total ordered quantity and Total delivered quantity

➤ Regional Analysis:

Orders were analyzed by region to identify regions with higher order volumes. Regional Analysis can help businesses understand demand distribution and improve resource and inventory planning.

**5. Insights:**

- Some products have significantly higher demand than others and may require better inventory planning.
- Differences between ordered and delivered quantities indicate incomplete order fulfillment.
- Suppliers handling many orders are important for maintaining supply chain continuity.
- Orders with important KPI for monitoring both delivery timelines and order completeness.
- OTIF is an important KPI for monitoring both delivery timelines and order completeness.
- Certain product categories may have higher order volumes or higher average prices than others.
- Comparing ordered quantity with delivered quantity helps identify supply shortages and fulfillment gaps.

**6. Conclusion:**

The Supply Chain Data Analysis project provides clear understanding of **orders, products, suppliers, quantities, regions, and delivery performance**.

Using **MYSQL** analysis, important operational metrics such as total orders, ordered quantity, delivery, quantity, supplier performance, product performance, and OTIF can be evaluated.

The analysis helps identify demand patterns, delivery gaps, supplier performance issues, and areas for operational improvement.

Overall, this project demonstrates how data analysis can support supply chain decisions-making by transforming raw operational data into meaningful business insights. The findings can help organizations improve **inventory planning, supplier management order fulfillment, delivery performance, and overall supply chain efficiency**.